Evaluating Expressions with Radicals

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	4 — Solving Problems with Rational Numbers
Lesson	Lesson 5 — Evaluating Expressions with Radicals	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.8.NSO.1.7

2. Standards

MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.

Must. Evaluate expressions containing square roots, cube roots and integer exponents, simplifying radicals by factoring within the benchmark's lists.

Must not. Perfect squares up to 225 and perfect cubes from −125 to 125, or fractions whose parts are each in those lists. Estimating a root that is not perfect was Unit 2. No fractional exponents.

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.5.1	Use patterns and structure to help understand and connect mathematical concepts.	Notes I — the radical bar is recognised as a grouping symbol by comparing √(9 + 16) with √9 + √16, so the rule is derived from the difference rather than announced.
MA.K12.MTR.6.1	Assess the reasonableness of solutions.	Boards 3 and 6 — students say whether a negative answer is possible at all before they evaluate, which is the whole square-root-versus-cube-root distinction.
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	Board 5 — when the room splits between 13 and 17 on √(25 + 144), the sixty-second re-vote before the reveal.

4. Learning Target and Essential Question

Learning target. I can evaluate an expression that contains square roots, cube roots and integer exponents.

Essential question. What does the radical bar do to the order in which an expression is evaluated?

Building on. Unit 2 (2.01–2.06) — perfect squares, perfect cubes and the principal square root; 3.04 — negative exponents; Grade 6 — the order of operations.

Working toward. 4.06 — the whole order of operations at once; 4.07–08 — real-world multi-step problems; MA.912.NSO.1.4 in Algebra 1.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	12	Teach from the front. Expand once per law; say the OFF line out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	11	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; finish tonight.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Why is $\sqrt{(9 + 16)}$ not the same as $\sqrt{9} + \sqrt{16}$, when $(9 + 16) \div 5$ is the same as $9 \div 5 + 16 \div 5$?	DOK 3
A cube root of a negative number is ordinary and a square root of one is not. What is it about the exponent 2 that makes the difference?	DOK 3
$\sqrt{49}$ is 7, but the equation $x^2 = 49$ has two solutions. Are those two statements in conflict?	DOK 3

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	$\sqrt{121}$	***11***	60.5 — the radicand was halved [Unit 2, 2.02: a root is not a half]; ±11 — the radical sign names the principal root only [8.NSO.1.7]
2	A square patio has an area of 169 square feet. Fencing for it costs 12 dollars a foot. How long is one side of the patio?	***13 feet***	anything built from 12 dollars — the unneeded figure was used [the question asks for a length, not a cost]; 52 — the perimeter, not a side; 84.5 — the area was halved
3	$\sqrt[3]{-125}$	***−5***	5 — the sign was dropped [8.NSO.1.7: an odd root keeps the sign of its radicand]; 'not a real number' — the square-root rule applied to a cube root [B1G-M 8.NSO.1.7 misconception: confusing square roots with cube roots]
4	$\sqrt{2^3 + 17}$	***5***	25 — the arithmetic under the bar was done and the root was not [8.NSO.1.7]; anything with two separate roots — the bar groups, so it is one radicand [8.NSO.1.7]
5	√(25 + 144). Which is the value?	***D — 13***	(A) the root of each piece was taken and the results added, 5 + 12 [8.NSO.1.7: the bar groups, so the radicand is 169]; (B) the sum was found and the root was never taken [8.NSO.1.7]; (C) the radicand was halved instead of rooted [Unit 2, 2.02: a root is not a half]
6	The cube root of −1/64. Which is the value?	***A — −1/4***	(B) the square-root rule for negatives was applied to a cube root [B1G-M 8.NSO.1.7 misconception]; (C) the square root of 1/64 was taken and the sign kept [B1G-M 8.NSO.1.7 misconception: confusing square roots with cube roots]; (D) the sign was dropped [8.NSO.1.7: an odd root keeps the sign of its radicand]
7	$\sqrt{\frac{4}{9}} + \sqrt[3]{\frac{8}{27}}$	***4/3***	12/36 — the roots were never taken, then 4/9 + 8/27 was added top-and-bottom [two errors: the roots, then Grade 5 fraction addition]; 2/3 — only one of the two radicals was evaluated
8	$\sqrt{5^3 - 25} \cdot 10^{-3}$	***1/100***	10,000 — 10^{-3} read as 10^3 [8.NSO.1.3: a negative exponent is a reciprocal]; −1/100 — the exponent's sign made the value's sign [B1G-M 8.NSO.1.3 task: −b vs b^{-1}]
9	Simon says that the cube root of −27 is not a real number, because you cannot take the root of a negative. Explain why he is wrong, and give the value.	***−3***	'−3' alone — the value without the reason [the item asks why he is wrong]; '3' — the sign dropped as well as the rule misread

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / IEP. Give the perfect-square and perfect-cube lists from Unit 2's reference sheet and let them stay out; a root is a lookup today, not a recall. On every board with arithmetic under the bar, the student draws a box around the radicand before writing anything, so the grouping is a physical act.

ELL. Radicand and radical are the two new nouns, and they are easy to confuse because they sound alike. Write √25 once with RADICAL circling the whole symbol and RADICAND labelling only the 25, and point at the labels on every board.

Enrichment. Ask for a pair of numbers for which √(a + b) really does equal √a + √b, and what has to be true of them. Then ask why the same question for a product has a different answer.

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the student has to say which root Simon's rule belongs to, not only that he is wrong, and give −3. Read the boards, not the papers.

Vocabulary. **radical**; **radicand**; **principal square root**

Materials. Whiteboards and markers, and Unit 2's reference sheet if students still want the perfect-square list. Calculators are allowed on everything; a calculator will refuse the cube root of a negative on some models, which is worth showing once and explaining rather than trusting.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, this skill continues tomorrow: one assignment, due at the start of the class after 4.06: Evaluate radical expressions — D8J.